

■ ■ Core Infrastructure & Framework Modules

D365FO Middleware Backend — System Infrastructure & Foundation Services

Target Audience: Backend Developers, System Administrators, DevOps Engineers.

1. Overview of Infrastructure Layer

The core infrastructure modules provide essential framework capabilities: security, authentication, D365FO OData connectivity, batch orchestration (CQRS), asynchronous queue management (BullMQ), resilience (circuit breakers & caching), database persistence, health monitoring, and logging.

2. Infrastructure Modules Breakdown

2.1 AuthModule & UserModule (src/modules/auth & src/modules/user)

Business Purpose & Safeguards

Secures API endpoints against unauthorized access, enforcing Role-Based Access Control (RBAC) for finance operators and system integration clients.

Key Components & Code Architecture

- **AuthService**: Handles user authentication, password verification using **Argon2** hashing, and JWT token issuance.
- **JwtAccessStrategy** & **JwtRefreshStrategy**: Passport strategies validating Bearer access tokens and refresh tokens.
- **RolesGuard** & **@Roles()** Decorator: Restricts administrative endpoints (e.g. batch cancellation, master data sync) to authorized roles.
- Mongoose Schema: **UserSchema** storing usernames, Argon2 password hashes, roles, active status, and refresh token hashes.

2.2 D365FOModule (src/modules/d365fo)

Business Purpose & D365FO Connection

Manages all direct communication with Microsoft Dynamics 365 Finance & Operations OData REST APIs.

Key Components & Code Architecture

- **D365FOAuthService**: Obtains and caches OAuth2 Bearer tokens from Azure Active Directory using Client Credentials Flow (**D365FO_CLIENT_ID**, **D365FO_CLIENT_SECRET**, **D365FO_RESOURCE**). Automatically refreshes tokens before expiry.
- **D365FOClientService**: HTTP wrapper around Axios featuring:
- **Opossum Circuit Breaker**: Stops outbound calls if D365FO returns consecutive 5xx errors, preventing system cascading failure.
- **Axios Retry**: Automatic exponential backoff retries for transient network drops (408, 502, 503, 504).
- **ODataQueryBuilderService**: Fluent query builder for constructing OData URL queries (**\$filter**, **\$select**, **\$expand**, **\$top**, **\$skip**).

2.3 DataBatchModule (src/modules/data-batch)

Business Purpose & CQRS Batch Management

Implements the Command Query Responsibility Segregation (CQRS) pattern to manage financial batch lifecycles. Every uploaded Excel sheet is converted into a **DataBatch** entity.

Key Components & Code Architecture

- **Commands & Handlers**:
- **CreateDataBatchCommand** / **CreateDataBatchHandler**: Creates batch records in MongoDB.
- **UpdateDataBatchStatusCommand** / **UpdateDataBatchStatusHandler**: Transitions batch state (**Pending** -> **Processing** -> **Completed** / **Failed**).
- **Queries & Handlers**:
- **GetDataBatchByIdQuery**, **GetPendingBatchesQuery**, **GetDataBatchErrorsQuery**.
- **Mongoose Schema (DataBatchSchema)**: Stores batch ID, processor type, total lines, status, creation date, D365FO voucher identifiers, and line-level error logs.

2.4 EntryProcessorModule (src/modules/entry-processor)

Business Purpose & Transformation Engine

Serves as the core data engine that transforms raw Excel rows into normalized, D365FO-compliant payload structures.

Key Components & Code Architecture

- **EntryProcessorBase**: Abstract base class providing common transformation methods:
- **parseDimensions()**: Splitting **ACCOUNTDISPLAYVALUE** into 15 financial dimension tags.
- **validateMonthIsolation()**: Enforcing single-month rules.
- **applyVoucherIntegrity()**: Ensuring lines with identical vouchers stay grouped.
- **EntryProcessorFactory**: Factory class maintaining a registry of all 28 entry processors, returning the appropriate processor based on **EntryProcessorTypes** enum.

2.5 QueueModule (src/modules/queue)

Business Purpose & Asynchronous Worker Distribution

Leverages **BullMQ** (powered by Redis) to process posting jobs in background threads without blocking API responsiveness.

Key Queues & Workers

1. **dfo-free-text-invoice-queue**: Handles AR Free Text Invoices.
2. **dfo-vendor-journal-queue**: Handles AP Vendor Invoice Journals.
3. **dfo-ledger-journal-queue**: Handles Month-End GL Adjustments & Custody Settlement.
4. **dfo-customer-payment-journal-queue**: Handles Customer Payment Receipts.
5. **master-data-sync-queue**: Handles background master data pre-fetching.

2.6 ResilienceModule & DBModule (src/modules/resilience & src/modules/db)

Business Purpose & High Availability

Ensures system stability under high load or ERP downtime.

- **ResilienceModule**: Provides multi-layer caching (L1 Memory, L2 Redis, L3 MongoDB) and circuit breaker monitoring.
- **DBModule**: Manages Mongoose connection pooling to MongoDB 8 (**MONGODB_URI**), handling reconnection logic.

2.7 ObservabilityModule, HealthModule, SchedulerModule & SettingsModule

- **ObservabilityModule**: Structured JSON logging powered by **Pino** (**nestjs-pino**), redacting sensitive fields (passwords, tokens, client secrets).
- **HealthModule**: Exposes **/health** probe endpoints checking MongoDB, Redis, and D365FO OData connectivity.
- **SchedulerModule**: Manages cron schedules for automated master data sync and log cleanup.
- **SettingsModule**: Stores dynamic application settings and posting thresholds in MongoDB.